



# Birla Institute of Technology & Science, Pilani

Pilani Campus

**ACADEMIC— UNDER GRADUATE STUDIES DIVISION**

**ACADEMIC— GRADUATE STUDIES & RESEARCH DIVISION**

**FIRST SEMESTER 2026-2027**

**Course Handout Part II**

Date:01/08/2026

In addition to part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

**Course No. : DE G513**

**Course Title : TRIBOLOGY**

**Instructor-in-charge : Dr. ARUN KUMAR JALAN**

## **1. Course Description:**

Introduction, lubricants and lubrication, surface texture, bearing materials, fundamentals of viscous flow, Reynolds equation and applications, thrust bearings, journal bearings, squeeze-film bearings, hydrostatic bearings, gas bearings, dry and starved bearings, selecting bearing type and size, principles and operating limits, friction, wear and lubrication.

## **2. Course Objectives:**

| No  | Course Objectives                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------|
| CO1 | Provide an introduction to lubricants and lubrication.                                                                    |
| CO2 | Selection and study of different bearings, i.e. Hydrodynamic, hydrostatic, elasto-hydrodynamic and antifriction bearings. |
| CO3 | Study on surface textures and its standardization.                                                                        |
| CO4 | Learn standard principles of friction, wear and lubrication.                                                              |

## **3. Learning Outcomes :**

| No  | Learning Outcomes                                             |
|-----|---------------------------------------------------------------|
| LO1 | An ability to select proper bearing to standardize the system |
| LO2 | Understanding importance of lubrication                       |
| LO3 | An ability to measure surface textures.                       |
| LO4 | Understanding importance of friction and wear.                |

## **4. Text Book:**

a) **TB1:** Basu, S. K; Ahuja, B. B., Sengupta, S. N., “Fundamentals of Tribology”, Prentice Hall of India 2005.

b) **TB2:** Class lecture notes



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## 5. Reference Books:

- RB1: Stachowiak G.W.; Batchelor A.W.; “Engineering Tribology”; Elsevier, 1993.
- RB2: Fuller D.D.; “Theory and Practice of Lubrication for Engineers”, John Wiley & Sons Inc. 1965.
- RB3: Bhushan, B. Introduction to Tribology, John Wiley & Sons Inc. 2nd Edition 2013.

## 6. Course Plan: -

| Lect. No | Topics to be covered                                     | Learning Objective                                                                  | Reference Chap./Sec           |
|----------|----------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------|
| 1-2      | Introduction to Tribology                                | Basic introduction of tribology and importance of friction, wear and lubricants     | TB 1: Ch.1                    |
| 3-5      | Viscosity of lubricants                                  | Study the Physical properties of Lubricants and Measurements                        | TB 1: Ch.2<br>RB 1: Ch.2      |
| 6-9      | Surface roughness and standardize measurement techniques | Understand surface textures: standards, Measurement and statistical analysis        | TB 1: Ch.3<br>RB 3: Ch.2      |
| 10-18    | Hydrodynamic bearings: Slider and Journal                | Study on hydrodynamic lubrication                                                   | TB 1: Ch.4 & 5<br>RB 1: Ch. 4 |
| 19-21    | Hydrostatic Bearings                                     | Study on hydrostatic lubrication                                                    | TB 1: Ch.7<br>RB 1: Ch. 6     |
| 22-24    | Elastohydrodynamic lubrication                           | Study on elastohydrodynamic lubrication                                             | TB 1: Ch.8                    |
| 25-30    | Friction and wear                                        | Understanding importance of friction and wear                                       | TB 1: Ch.9<br>RB 3: Ch 5 & 7  |
| 31-34    | Antifriction Bearing                                     | Study on antifriction bearing                                                       | TB 1: Ch.10<br>TB 2           |
| 35-36    | Lubricants                                               | Different types of Lubricants, composition, Selection of lubricants and their uses. | TB 1: Ch.12<br>RB 1: Ch. 3    |

## 7. Evaluation Scheme:

| Component                       | Duration | Weightage (%) | Date & Time | Remarks |
|---------------------------------|----------|---------------|-------------|---------|
| Mid-sem                         | 90 min   | 25            | <TEST 1>    | CB+OB   |
| Seminar/ Project/Lab assignment |          | 30            |             | OB      |
| Quiz/Test                       |          | 10            |             | OB/CB   |
| Compre. Exam                    | 3 hrs    | 35            | <TEST C>    | OB+CB   |

**8. Consultation Hours: Chamber no 2229, Wednesday 4.00 pm to 5.30 pm**

**9. Notice:** All the notices regarding the course will be displayed on the Nalanda or Mechanical Engineering Department Notice Board

**Instructor-in-charge  
DE G513**



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